

Operations on Functions

Algebra Worksheet · Grade 9–11

Name: _____

Date: _____

Learning Objectives

- Add and subtract functions by combining like terms
- Multiply functions and expand polynomial products
- Divide functions and simplify rational expressions

Problems

1. Given $f(x) = 2x + 5$ and $g(x) = x + 3$, find $(f + g)(x)$.

$$f(x) = 2x + 5, \quad g(x) = x + 3$$

2. Given $f(x) = 3x + 8$ and $g(x) = 2x - 12$, find $(f + g)(x)$.

$$f(x) = 3x + 8, \quad g(x) = 2x - 12$$

3. Given $f(x) = 3x + 8$ and $g(x) = 2x - 12$, find $(g - f)(x)$.

$$f(x) = 3x + 8, \quad g(x) = 2x - 12$$

4. Given $f(x) = 4x - 3$ and $g(x) = x + 7$, find $(f - g)(x)$.

$$f(x) = 4x - 3, \quad g(x) = x + 7$$

5. Given $f(x) = x + 1$ and $g(x) = x - 8$, find $(f \text{ times } g)(x)$ by expanding the product.

$$f(x) = x + 1, \quad g(x) = x - 8$$

6. Given $f(x) = x + 1$ and $g(x) = x^2 - 8x + 7$, find $(f \text{ divided by } g)(x)$ and simplify. State any restrictions on x .

Scan to watch



$$f(x) = x + 1, \quad g(x) = x^2 - 8x + 7$$

7. Given $f(x) = x$ squared minus 1 and $g(x) = x + 1$, find $(f \text{ divided by } g)(x)$ and simplify. State any restrictions on x .

$$f(x) = x^2 - 1, \quad g(x) = x + 1$$

8. Given $f(x) = x$ squared plus $3x + 2$ and $g(x) = x + 2$, find $(f \text{ divided by } g)(x)$ and simplify. State any restrictions on x .

$$f(x) = x^2 + 3x + 2, \quad g(x) = x + 2$$

9. Given $f(x) = x$ squared minus x minus 2 and $g(x) = x + 1$, find $(f \text{ times } g)(x)$ by expanding the product fully.

$$f(x) = x^2 - x - 2, \quad g(x) = x + 1$$

10. Given $f(x) = x$ cubed plus x squared minus $8x$ minus 8 and $g(x) = x$ squared minus 4, find $(f \text{ divided by } g)(x)$, simplify fully, and state all restrictions on x .

$$f(x) = x^3 + x^2 - 8x - 8, \quad g(x) = x^2 - 4$$

Scan to watch



Operations on Functions — Answer Key

Algebra Worksheet · Grade 9–11

Answer Key

1. Answer: $(f + g)(x) = 3x + 8$

- Write $(f + g)(x) = f(x) + g(x) = (2x + 5) + (x + 3)$
 - Combine like terms: $2x + x = 3x$ and $5 + 3 = 8$
 - $(f + g)(x) = 3x + 8$
-

2. Answer: $(f + g)(x) = 5x - 4$

- Write $(f + g)(x) = (3x + 8) + (2x - 12)$
 - Combine like terms: $3x + 2x = 5x$ and $8 - 12 = -4$
 - $(f + g)(x) = 5x - 4$
-

3. Answer: $(g - f)(x) = -x - 20$

- Write $(g - f)(x) = g(x) - f(x) = (2x - 12) - (3x + 8)$
 - Distribute the negative sign: $2x - 12 - 3x - 8$
 - Combine like terms: $2x - 3x = -x$ and $-12 - 8 = -20$
 - $(g - f)(x) = -x - 20$
-

4. Answer: $(f - g)(x) = 3x - 10$

- Write $(f - g)(x) = f(x) - g(x) = (4x - 3) - (x + 7)$
 - Distribute the negative sign: $4x - 3 - x - 7$
 - Combine like terms: $4x - x = 3x$ and $-3 - 7 = -10$
 - $(f - g)(x) = 3x - 10$
-

5. Answer: $(f \cdot g)(x) = x^2 - 7x - 8$

- Write $(f \cdot g)(x) = (x + 1)(x - 8)$
 - Use FOIL: $x \cdot x + x \cdot (-8) + 1 \cdot x + 1 \cdot (-8)$
 - Simplify: $x^2 - 8x + x - 8$
 - $(f \cdot g)(x) = x^2 - 7x - 8$
-

6. Answer: $(f/g)(x) = 1/(x - 7)$, $x \neq -1$, $x \neq 7$

- Write $(f/g)(x) = (x + 1) / (x^2 - 8x + 7)$
 - Factor the denominator: $x^2 - 8x + 7 = (x - 1)(x - 7)$
 - Rewrite: $(x + 1) / [(x - 1)(x - 7)]$
 - No common factors cancel; domain restrictions: $x \neq 1$ and $x \neq 7$
 - Wait — check: denominator factors are $(x - 1)(x - 7)$; numerator is $(x + 1)$, no cancellation
 - $(f/g)(x) = (x + 1) / [(x - 1)(x - 7)]$, $x \neq 1$, $x \neq 7$
-

7. Answer: $(f/g)(x) = x - 1$, $x \neq -1$

- Write $(f/g)(x) = (x^2 - 1) / (x + 1)$
- Factor the numerator using difference of squares: $x^2 - 1 = (x + 1)(x - 1)$

Scan to watch





- Cancel the common factor $(x + 1)$: $(x + 1)(x - 1) / (x + 1) = x - 1$
- Restriction: $x \neq -1$ (denominator cannot equal zero)
- $(f/g)(x) = x - 1, x \neq -1$

8. Answer: $(f/g)(x) = x + 1, x \neq -2$

- Write $(f/g)(x) = (x^2 + 3x + 2) / (x + 2)$
- Factor the numerator: $x^2 + 3x + 2 = (x + 2)(x + 1)$
- Cancel the common factor $(x + 2)$: $(x + 2)(x + 1) / (x + 2) = x + 1$
- Restriction: $x \neq -2$
- $(f/g)(x) = x + 1, x \neq -2$

9. Answer: $(f \cdot g)(x) = x^3 - 3x - 2$

- Write $(f \cdot g)(x) = (x^2 - x - 2)(x + 1)$
- Distribute: $x^2 \cdot x + x^2 \cdot 1 + (-x) \cdot x + (-x) \cdot 1 + (-2) \cdot x + (-2) \cdot 1$
- Expand: $x^3 + x^2 - x^2 - x - 2x - 2$
- Combine like terms: $x^3 + (x^2 - x^2) + (-x - 2x) - 2$
- $(f \cdot g)(x) = x^3 - 3x - 2$

10. Answer: $(f/g)(x) = (x + 1)/(x - 2) \cdot (x+2)/(x+2)$ simplifies to $(x^2 + x) / (x - 2)$ — see steps; final: $(x+1)(x+2)/[(x-2)(x+2)] = (x+1)/(x-2), x \neq \pm 2$

- Factor the numerator by grouping: $x^3 + x^2 - 8x - 8 = x^2(x + 1) - 8(x + 1) = (x^2 - 8)(x + 1)$... try again: group as $(x^3 + x^2) + (-8x - 8) = x^2(x + 1) - 8(x + 1) = (x + 1)(x^2 - 8)$
- Factor the denominator: $x^2 - 4 = (x + 2)(x - 2)$
- Write $(f/g)(x) = (x + 1)(x^2 - 8) / [(x + 2)(x - 2)]$
- Check for common factors: $x^2 - 8$ does not factor to include $(x+2)$ or $(x-2)$, so no cancellation
- Restrictions: $x \neq 2$ and $x \neq -2$
- $(f/g)(x) = [(x + 1)(x^2 - 8)] / [(x + 2)(x - 2)], x \neq \pm 2$

Scan to watch

